

Non-singular fast terminal sliding mode control of permanent magnet synchronous motor for automatic guided vehicle based on disturbance compensation

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Abstract

In the smart factory, the loading and unloading of goods by automatic guided vehicle in the driving state will cause speed fluctuations. To this problem, this article takes the permanent magnet synchronous motor of automatic guided vehicle as the research object and studies the motor speed control of automatic guided vehicle when the load changes. This article presents a non-singular fast terminal sliding mode control strategy for permanent magnet synchronous motor based on disturbance feed-forward compensation. In order to solve the problem of chattering and slow reaching the speed of sliding mode control, this article designs an adaptive exponential reaching law, which can not only suppress sliding mode chattering but also adjust the reaching speed adaptively according to the system state. In order to solve the problem of external load disturbance, this article designs a sliding mode disturbance observer, which suppresses the influence of load disturbance on speed by compensating the observed values feed-forward into the current loop. Compared with proportional–integral and sliding mode control, the control strategy adopted in this article reduces the speed fluctuation by 3.12% and 2.00%, and the response time is reduced by 67.9% and 33.3%, respectively—finally, the load mutation experiments of automatic guided vehicle with permanent magnet synchronous motor as the driving motor is carried out. The experimental results further verify the effectiveness of the control strategy. Simulation and experimental results show that the proposed control strategy has better anti-interference ability and faster response speed.

Keywords

Automatic guided vehicle, permanent magnet synchronous motor, load mutation, non-singular fast terminal sliding mode control, sliding mode disturbance observer

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Introduction

Automatic guided vehicle (AGV), as a highly intelligent and automated transportation equipment, is widely used due to its advantages of high intelligence, strong mobility, and high control accuracy.^{1,2} The precise control of AGV speed guarantees stable and precise operation, and achieving precise speed control depends on the control performance of the drive motor speed control system. Permanent magnet synchronous motor (PMSM) has the advantages of small size, simple structure, high power density, and high control accuracy.³ It is suitable for driving motors of AGV, and its performance directly affects the performance of AGV. It is suitable for the drive motor of AGV, and its performance directly affects the performance of AGV. With

the continuous enrichment of the working scene of AGV, the performance requirements of PMSM are getting higher and higher. When AGV is working, the load will change suddenly due to the loading and unloading of goods, which will cause the load of the driving motor of the AGV to change suddenly. This mutation will cause the speed fluctuation of the PMSM, which will

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affect the regular operation of the AGV. Therefore, to ensure the stability of the speed of the AGV when the load changes suddenly, it is essential to study the anti-interference performance of the PMSM.

In order to suppress the influence of load disturbance on PMSM, domestic and foreign scholars have conducted much research on the control of PMSM speed control systems. The proportional–integral (PI) control method is a commonly used control method for the PMSM speed control system, but this method is sensitive to external interference. When the internal parameters of the system change or the load changes, the speed of the PMSM changes significantly, the PI control method can only meet the control requirements within a specific range and cannot effectively suppress the influence of load mutation on the speed control system. With further research, some modern control theory is applied to the PMSM speed control system, such as fuzzy control,⁴ adaptive control,⁵ neural networks,⁶ active disturbance rejection control,⁷ model predictive control,⁸ and sliding mode control (SMC).⁹ SMC, because of its simple algorithm, fast response speed, low requirements for model accuracy, and strong robustness to parameter perturbation and external disturbance, especially for system control, has become a research hotspot in the current speed control of PMSM.

Domestic and foreign scholars have conducted in-depth research on the sliding mode control of PMSM. The research hotspots mainly include sliding mode surface design, reaching law design, and multi-theory combined control research. The aim is to improve the sliding mode control's response speed and suppress the system's chattering and disturbance rejection performance. In order to improve the effect of SMC, Xu et al.¹⁰ used a nonlinear terminal sliding mode surface to improve the anti-interference performance and the system's convergence speed. However, the terminal sliding surface has a singular value problem. Wang et al.¹¹ proposed a non-singular terminal sliding mode controller to avoid the singular value problem. Liu et al.¹² designed a non-singular fast terminal sliding mode surface based on the terminal sliding mode, which not only avoids the singular value problem but also improves the convergence speed of the system.

The reaching law determines the reaching speed of the system state point to the sliding mode surface. The quality of the reaching law directly affects the reaching speed and chattering of the SMC system. The commonly used reaching laws are the constant speed reaching law, exponential reaching law, and the power reaching law. However, these reaching laws need to be balanced regarding convergence speed and chattering suppression. In order to solve this problem, Wang and Wei¹³ proposed an improved exponential law. In order to overcome the contradiction between the fast response speed and heavy chattering of the conventional non-singular fast terminal sliding mode control (NFTSMC), Xu et al.¹⁴ proposed a non-singular fast

terminal sliding mode controller with improved reaching law. Qu et al.¹⁵ proposed a fast variable sliding mode reaching law, which suppresses the system's chattering and improves the system's response speed. Sun et al.¹⁶ proposed a hybrid reaching law, which balances both the reaching speed and the chattering problem and has good dynamic characteristics.

In addition to improving the anti-interference ability of the controller, another effective method is to estimate the load disturbance through the disturbance observer and feed-forward the observation value to the current loop to suppress the influence of the load disturbance on the PMSM.¹⁷ Zhang et al.¹⁸ used the extended sliding mode disturbance observer (SMDO) to estimate the system disturbance. They successfully suppressed external disturbance using the composite control method of sliding mode control and disturbance observer. Nguyen et al.¹⁹ proposed a nonlinear disturbance observer (NDOB) with anti-interference ability, which can accurately estimate the uncertain disturbance and automatically adjust the convergence speed of the observer. Wu and Li²⁰ used the sliding mode state observer to estimate the total disturbance affecting the system performance and proposed an adaptive disturbance compensation finite control set optimization control strategy. Based on the prescribed performance SMC, Dai et al.²¹ proposed a new disturbance observer to estimate the load disturbance in actual operation, which improved the anti-interference ability of PMSM. Yang et al.²² proposed a new disturbance observer based on an iterative learning strategy, which enhances the conventional observer using iterative learning law to improve the estimation ability of periodic disturbance.

In this article, considering the above problem and to further improve the dynamic performance of the PMSM speed control system, an NFTSMC based on a novel reaching law is proposed. The new reaching law introduces a system state variable and a sliding mode surface on the basis of the traditional exponential reaching law so that the convergence speed can be adaptively adjusted according to the system state. Then, considering the effect of external load perturbations on the PMSM, an SMDO is designed to observe the disturbance and feed-forward the observed values into the compensated speed controller. A composite control strategy combining NFTSMC and SMDO is developed. Simulation and experimental results show that the control strategy has a faster dynamic response speed and better anti-interference performance. The main contributions of this article are as follows: (1) a speed controller based on a new sliding mode reaching law has been designed for PMSM, which can suppress sliding mode chattering and achieve a faster dynamic response. (2) The designed SMDO is able to estimate the external disturbances and suppresses the disturbance of the load disturbance on the speed control system of the PMSM by observing and compensating for the load disturbance. (3) A composite control method is

composed of the NFTSMC and the SMDO to improve the dynamic response speed and anti-interference performance of PMSM.

The rest of this article is arranged as follows. The mathematical model of PMSM is described in section “Mathematical model of PMSM.” The speed controller is designed in section “Design of NFTSMC speed controller.” We design an SMDO in section “Design of SMDO.” Simulation and experiments analysis are shown in the section “Simulation and experimental verification.” Conclusions are provided in section “Conclusion.”

Mathematical model of PMSM

Three-phase PMSM is a strong coupling, a complex nonlinear system with multivariable parameters. In order to simplify the analysis, when establishing the mathematical model of the motor, it is assumed that the motor runs under the following conditions: the stator core is unsaturated, the hysteresis and eddy current loss of the motor is not considered, the spatial magnetic field is sinusoidal, and there is no damping disturbance on the rotor. Based on the above assumptions, the voltage equation of PMSM in the d - q axis coordinate system can be expressed as

$$\begin{cases} u_d = Ri_d + L_d \frac{di_d}{dt} - L_q \omega_e i_q \\ u_q = Ri_q + \omega_e \psi_f + \omega_e L_d i_d + L_q \frac{di_q}{dt} \end{cases} \quad (1)$$

The electromagnetic torque equation of PMSM is

$$T_e = 1.5p_n[i_q \psi_f + (L_d - L_q)i_d i_q] \quad (2)$$

For surface-mounted PMSM, $L_d = L_q$. Therefore, the torque equation can be simplified as follows

$$T_e = 1.5p_n i_q \psi_f \quad (3)$$

The motion equation of PMSM is

$$T_e - T_L = B\omega_m + J \frac{d\omega_m}{dt} \quad (4)$$

In the speed control system of PMSM, there exists external load disturbance. Therefore, when considering load disturbance, the motion equation of PMSM can be expressed as

$$T_e - T_L - T_L^* = B\omega_m + J_m \frac{d\omega_m}{dt} \quad (5)$$

Due to compensating the load disturbance as a feed-forward to the speed controller, the load disturbance is not considered in the design of the speed controller.

Combining equations (3) and (4), the motion equation of PMSM can be converted as follows

$$\frac{d\omega_m}{dt} = \frac{3p_n \psi_f}{2J} i_q - \frac{T_L}{J} - \frac{B}{J} \omega_m \quad (6)$$

Design of NFTSMC speed controller

Design of NFTSMC

The state variables of the PMSM system are defined as follows

$$\begin{cases} x_1 = \omega - \omega_m \\ x_2 = \dot{x}_1 = -\dot{\omega}_m \end{cases} \quad (7)$$

Combining equations (6) and (7) can obtain

$$\begin{cases} \dot{x}_1 = -\frac{d\omega_m}{dt} = -\frac{3p_n \psi_f}{2J} i_q + \frac{T_L}{J} + \frac{B}{J} \omega_m \\ \dot{x}_2 = -\frac{d\dot{\omega}_m}{dt} = -\frac{3p_n \psi_f}{2J} \dot{i}_q - \frac{B}{J} x_2 \end{cases} \quad (8)$$

Therefore, the state equation of the PMSM system is as follows

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{B}{J} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{3p_n \psi_f}{2J} \end{bmatrix} \dot{i}_q \quad (9)$$

Construct a non-singular fast terminal sliding surface as follows

$$s = x_1 + \alpha x_1^{g/h} + \beta x_2^{p/q} \quad (10)$$

where $\alpha > 0$, $\beta > 0$, g, h, p , and q are positive odd numbers, and $1 < p/q < 2$, $g/h > p/q$. Taking the derivative of equation (10) yields

$$\begin{aligned} \dot{s} &= \dot{x}_1 + \frac{\alpha g}{h} x_1^{\frac{g}{h}-1} \dot{x}_1 + \frac{\beta p}{q} x_2^{\frac{p}{q}-1} \dot{x}_2 \\ &= x_2 + \frac{\alpha g}{h} x_1^{\frac{g}{h}-1} x_2 + \frac{\beta p}{q} x_2^{\frac{p}{q}-1} \dot{x}_2 \end{aligned} \quad (11)$$

According to equation (11), there is no negative exponential power term for the first-order derivative of the sliding mode surface, avoiding singularity problems.

The traditional exponential reaching law function is as follows

$$\dot{s} = -b \operatorname{sgn}(s) - cs, \varepsilon > 0, \quad k > 0 \quad (12)$$

where $\operatorname{sgn}(\cdot)$ is the sign function, $b \operatorname{sgn}(s)$ is the constant velocity reaching term, and cs is the pure exponential term.

The traditional exponential reaching law can improve the response speed and robustness of the system to a certain extent, but it also causes problems such as system chattering. Due to discontinuous function

$\text{sgn}(s)$, when the system state s is close to the sliding surface, the exponential reaching law is switched to a small neighborhood of the sliding surface. When the state trajectory moves to the origin, it cannot approach the origin, but produces chattering near it. This chattering may excite high-frequency components not considered in system modeling and increase the burden on the controller. In order to overcome the shortcomings of the traditional exponential reaching law, this article introduces the system variables into the reaching law based on the exponential reaching law. An adaptive exponential reaching law is proposed

$$\dot{s} = -\varepsilon(1 - e^{-\delta|s|})\text{sgn}(s) - k|x_1|^\lambda s \quad (13)$$

where $\varepsilon, k, d, l > 0$, $\lim_{t \rightarrow \infty} |x_1| = 0$.

According to equation (13), it can be seen that the proposed adaptive exponential reaching law introduces the sliding mode surface s and the system state variable x_1 so that the reaching speed can be adjusted adaptively with the change of the system state. When the system state is far away from the sliding surface, $|s|$ is larger, so $1 - e^{-\delta|s|}$ approaches 1. The exponential reaching part $k|x_1|^\lambda s$ is greater than ks , which improves the convergence speed when the system state is far from the sliding mode surface. When $|s|$ approaches 0, the exponential part gradually approaches $0.1 - e^{-\delta|s|}$ will gradually approach 0. The speed of the system state gradually approaches 0 to achieve the purpose of suppressing chattering. Therefore, the new adaptive exponential reaching law can not only suppress chattering but also has a faster-reaching speed.

In order to better improve the effect of SMC, the reaching law is further improved. The sign function has the problem of input discontinuity to a certain extent, and frequent switching will cause system chattering. The sigmoid function is relatively smooth. Using the sigmoid function instead of the sign function, the chattering problem of the system is reduced, and the dynamic quality of the approaching process is improved. The $\text{sgn}(s)$ and $\text{sigmoid}(s)$ function curves are shown in Figure 1.

The improved approach law is as follows

$$\dot{s} = -\varepsilon(1 - e^{-\delta|s|})\text{sigmoid}(s) - k|x_1|^\lambda s \quad (14)$$

The $\text{sigmoid}(s)$ function is as follows

$$\text{sigmoid}(s) = \frac{2}{1 + e^{-as}} - 1 \quad (15)$$

where $a > 0$ is used to adjust the slope of the $\text{sigmoid}(s)$ function.

The expression of the speed controller obtained in the combination equations (6), (7), and (11) is as follows

$$i_q = \frac{2J}{3p_n\psi_f} \int_0^t \left[\frac{q}{\beta p} x_2^{2-p/q} \left(1 + \frac{\alpha g}{h} x_1^{g/h-1} \right) - \frac{B}{J} x_2 + \varepsilon(1 - e^{-\delta|s|})\text{sigmoid}(s) + k|x_1|^\lambda s \right] dt \quad (16)$$

Stability analysis of speed controller

The stability of the designed speed controller is verified according to the Lyapunov function, and the Lyapunov function is constructed

$$V = \frac{1}{2} s^2 \quad (17)$$

Derivation of equation (16)

$$\dot{V} = s\dot{s} \quad (18)$$

According to the Lyapunov stability theory, the reachability condition for sliding mode motion is

$$\dot{V} = s\dot{s} \leq 0 \quad (19)$$

Combining equations (10), (13), and (18), the following expressions can be obtained

$$\begin{aligned} \dot{V} &= s\dot{s} \\ &= s \left[-\varepsilon(1 - e^{-\delta|s|})\text{sigmoid}(s) - k|x_1|^\lambda s \right] \\ &= -\varepsilon(1 - e^{-\delta|s|})|s \cdot \text{sigmoid}(s)| - k|x_1|^\lambda s^2 \\ &\leq 0 \end{aligned} \quad (20)$$

According to the Lyapunov stability theory, the designed speed controller satisfies the stability theory. It is proved that the speed controller can make the system error converge quickly and meet the sliding mode arrival condition.

Design of SMDO

NFTSMC can reduce the influence of external disturbance on the system. However, for the sizable external disturbance, the anti-disturbance ability of SMC is not enough. In order to better suppress the influence of external disturbance on the speed, the SMDO is used to estimate the disturbance of the system in real time, and the observation value is converted into compensation current, which is feed-forward compensated to the

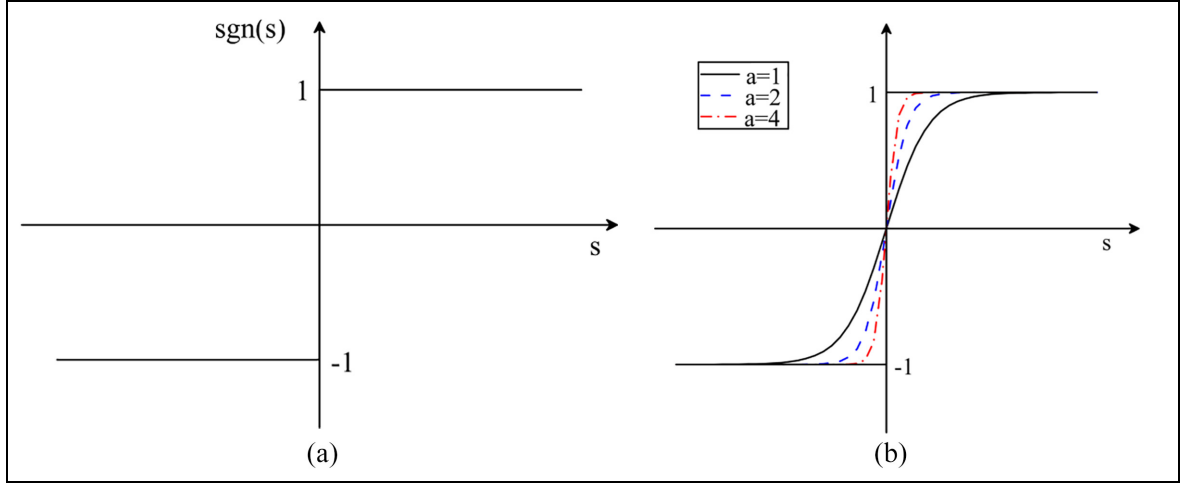


Figure 1. Function curve of (a) $\text{sgn}(s)$ and (b) $\text{sigmoid}(s)$.

current loop to enhance the ability of the system to resist load disturbance.

In the PMSM speed control system, the mechanical cycle of the motor system is much larger than the electrical cycle. In a sampling period, the load torque can be approximately unchanged. Therefore, the first derivative of the load torque is 0, that is, $\dot{T}_L = 0$. Construct the state equation by combining equation (6) with $\dot{T}_L$

$$\begin{cases} \dot{\omega}_m = \frac{3p_n\psi_f}{2J}i_q - \frac{T_f}{J} - \frac{B}{J}\omega_m \\ \dot{T}_L = 0 \end{cases} \quad (21)$$

Based on equation (21), the load torque T_L and the motor speed ω_m are taken as the observation objects, and the mathematical model of the SMDO is established

$$\begin{cases} \dot{\omega}_m^* = \frac{3p_n\psi_f}{2J}i_q - \frac{T_L^*}{J} - \frac{B}{J}\omega_m^* + U \\ \dot{T}_L^* = k_1 U \end{cases} \quad (22)$$

where $U = k_2 \text{sgn}(\omega_m^* - \omega_m)$, k_1 is the feedback gain and k_2 is the sliding mode gain.

The observation error equation of the SMDO is obtained by subtracting equation (22) from equation (21)

$$\begin{cases} \dot{e}_1 = -\frac{1}{J}e_2 - \frac{B}{J}e_1 + U \\ \dot{e}_2 = k_1 U \end{cases} \quad (23)$$

where $e_1 = \omega_m^* - \omega_m$ is the speed estimation error, and $e_2 = T_L^* - T_L$ is the load torque error of the motor

$$s_1 = e_1 = \omega_m^* - \omega_m \quad (24)$$

When the observer state enters the sliding mode surface, $s_1 = \dot{s}_1 = 0$ and the angular velocity error $e_1 = \dot{e}_1 = 0$. And by incorporating it into equation (23), it can be obtained that

$$\frac{1}{J}e_2 = U \quad (25)$$

Because $e_2 = T_L^* - T_L$, it can be obtained from equation 25

$$T_L^* = JU - T_L \quad (26)$$

Equations (3) and (4) can be obtained as follows

$$T_L = \frac{3}{2}p_n i_q \psi_f - B\omega_m - J_m \frac{d\omega_m}{dt} \quad (27)$$

According to equations (26) and (27), it can be obtained that

$$T_L^* = JU - \frac{3}{2}p_n i_q \psi_f + B\omega_m + J_m \frac{d\omega_m}{dt} \quad (28)$$

In order to analyze the stability of the sliding mode disturbance observer, the Lyapunov function is constructed as follows

$$V_1 = \frac{1}{2}s_1^2 \quad (29)$$

Derivation of equation (29)

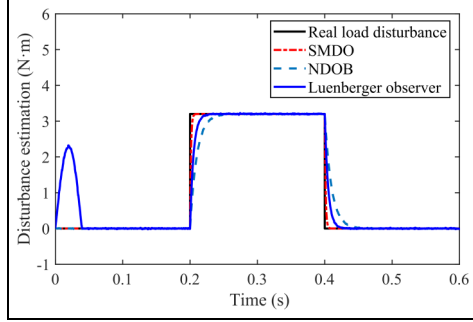
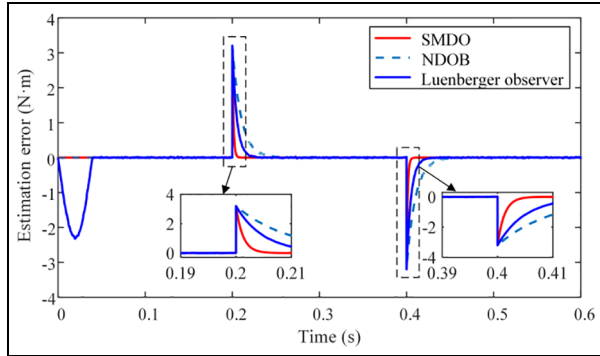
$$\begin{aligned} \dot{V}_1 &= s_1 \dot{s}_1 = e_1 \left(-\frac{1}{J}e_2 - \frac{B}{J}e_1 + k_2 \text{sgn } e_1 \right) \\ &= -\frac{1}{J}e_1 e_2 - \frac{B}{J}e_1^2 + k_2 e_1 \text{sgn } e_1 \end{aligned} \quad (30)$$

According to the reachability condition of sliding mode control, $\dot{V}_1 = s_1 \dot{s}_1 \leq 0$, so it can be obtained that

$$-\frac{1}{J}e_1 e_2 - \frac{B}{J}e_1^2 + k_2 e_1 \text{sgn } e_1 \leq 0 \quad (31)$$

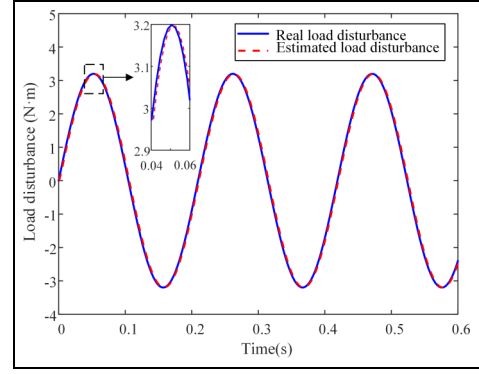
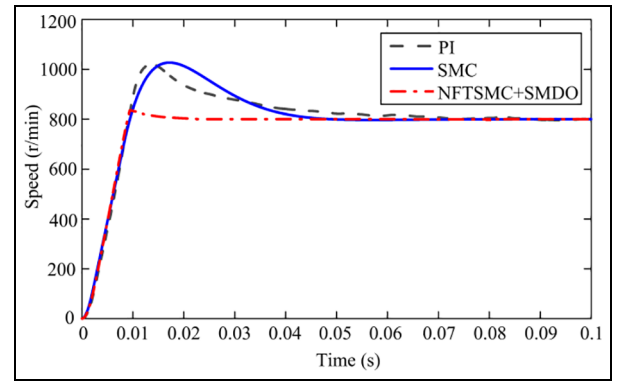
Table 1. PMSM parameters.

Parameter	Value
Rated power (W)	750
Rated power (r/min)	3000
Stator resistance (Ω)	0.085
Friction coefficient (N m s)	0.008
Stator inductance (mH)	4.5
Permanent-magnet flux linkage (Wb)	0.175
Rotary inertia (kg m^2)	0.003
Number of poles	4

**Figure 4.** Disturbance estimation of SMDO.**Figure 5.** Estimation error of SMDO.

seen from the figure that when the load mutation occurs, the observation error of the SMDO can converge to 0 within 6 ms. Compared with the SMDO, the Luenberger observer and NDOB, the SMDO used in this article can quickly and accurately estimate load disturbances. In order to verify the observation ability of the SMDO to the time-varying disturbance, the load disturbance of $T_L = 3.2\sin(30t)$ was observed. Figure 6 shows that the sliding mode disturbance observer proposed in this article can also quickly and accurately estimate time-varying disturbances.

Figure 7 shows the simulation results of the no-load start-up phase of the PMSM with different control strategies, and the given speed is 800 r/min. Figure 7 shows that the speed overshoot when using PI control is 27.4%, and the time to return to the given speed is about 9.3 ms. The overshoot of SMC control is 28.5%

**Figure 6.** Time-varying load torque.**Figure 7.** Speed response curve at start-up.

at start-up, and the time to return to the given speed is about 4.6 ms. When the control strategy of this article is adopted, the speed overshoot is 4.3%, and the time to return to the given speed is about 2.5 ms. The comparison shows that the control strategy proposed in this article has a faster response speed and a minor speed overshoot when the PMSM starts.

In order to verify the anti-interference performance of the PMSM speed control system in case of sudden load changes, simulation verification of sudden increase and decrease in load disturbances was conducted.

The load disturbance is shown in Figure 4. Figures 8 and 9 show the speed and torque response of the PMSM when the load changes.

Figure 8 shows the simulation results of the speed and torque of the PMSM when the load suddenly increases. The given speed of the PMSM is 800 r/min and the load torque increases from 0 to 3.2 N m at 0.2 s. Figure 8(a) shows that when the PI control method is used, the speed of PMSM decreases by a maximum of 42 r/min, and it takes 106 ms to recover to a given speed. When the SMC method is used, the speed drops by 33 r/min, and it takes 48 ms to recover to a given speed. Using the NFTSMC based on disturbance feed-forward compensation proposed in this article, the speed drop is 17 r/min, and the shortest time to return to the given speed is 32 ms. Compared to PI and SMC,

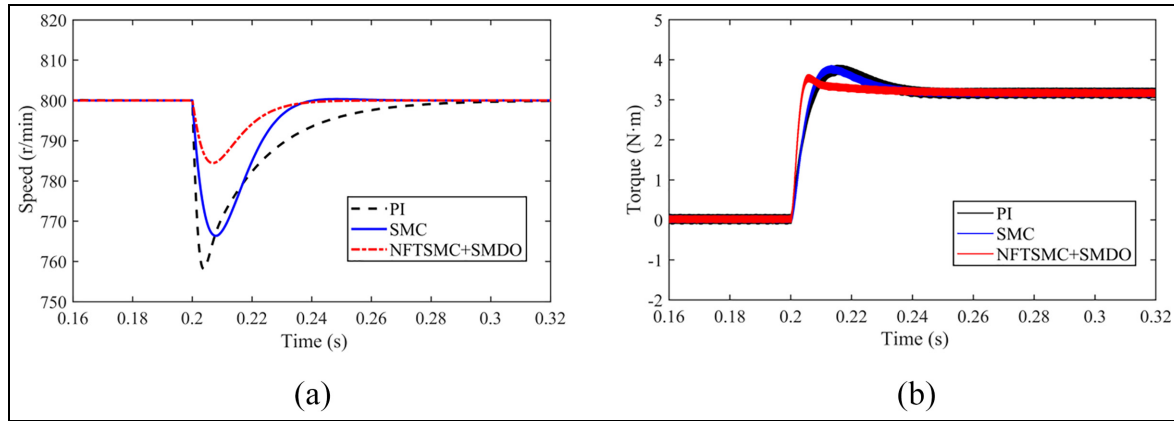


Figure 8. Simulation results of sudden load increase at 800 r/min. (a) Speed. (b) Torque.

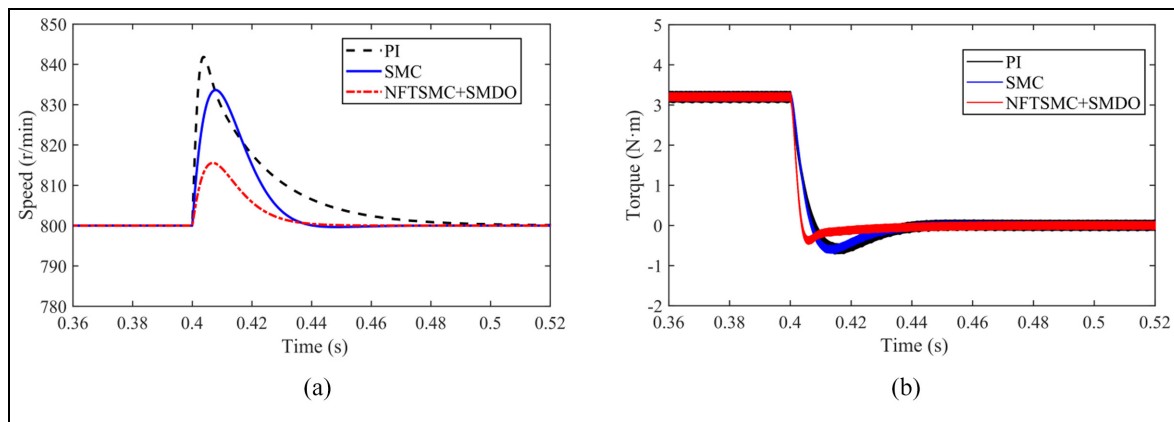


Figure 9. Simulation results of sudden load reduction at 800 r/min. (a) Speed. (b) Torque.

the control strategy proposed in this article reduces the speed drop by 3.12% and 2.00%. The recovery time to the given speed was reduced by 67.9 % and 33.3 %. Figure 8(b) shows that compared to PI and SMC, the method proposed in this article has a faster torque response and minimal torque fluctuation when the load suddenly increases. The results show that the proposed control strategy has a faster response speed and better anti-interference ability than PI and SMC when the load suddenly increases.

Figure 9 shows the simulation results of the speed and torque of the PMSM when the load suddenly decreases. The given speed of the PMSM is 800 r/min, and the load torque increases from 3.2 to 0 N·m at 0.4 s. Figure 9(a) shows that when the load suddenly decreases, the NFTSMC speed based on disturbance feed-forward compensation increases the least, and the recovery time to the given speed is the shortest. Figure 9(b) shows that compared to PI and SMC, the method proposed in this article has a faster torque response and minimal torque fluctuation when the load suddenly decreases. The results show that the control strategy proposed in this article has a faster response speed and better anti-interference ability than PI and SMC when the torque suddenly decreases.

However, in order to further verify the control performance of the control strategy at low speeds of the permanent magnet synchronous motor, we continued to compare the performance of the permanent magnet synchronous motor under different control methods at 300 r/min. Figures 10 and 11 show the speed response and torque response of a permanent magnet synchronous motor under sudden load changes at 300 r/min.

Figure 10 shows the simulation results of the speed and torque of the PMSM when the load suddenly increases at 300 r/min. The load torque increases from 0 to 3.2 N·m at 0.2 s. Figure 10(a) shows that when the load suddenly increases, the speed fluctuation of NFTSMC + SMDO is the smallest, and the time to recover to the given speed is the shortest. Figure 10(b) shows that the method proposed in this article has a faster torque response and more minor torque fluctuations when the load suddenly increases. Compared with Figure 8, it can be seen that the method proposed in this article has a smaller speed decrease and shorter recovery time at 300 r/min. The simulation results show that the control strategy proposed in this article also has a good control effect at low speed.

Figure 11 shows the simulation results of the speed and torque of the PMSM when the load suddenly

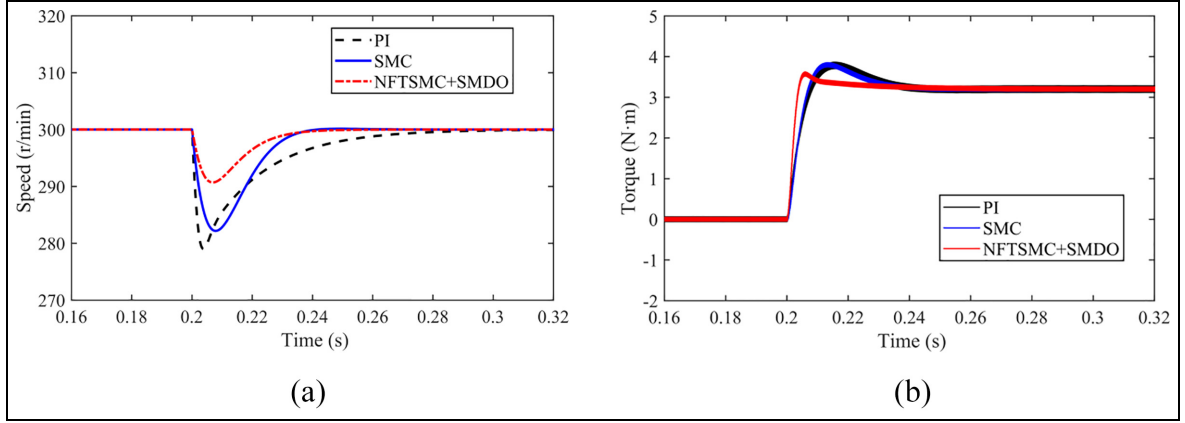


Figure 10. Simulation results of sudden load increase at 300 r/min. (a) Speed. (b) Torque.

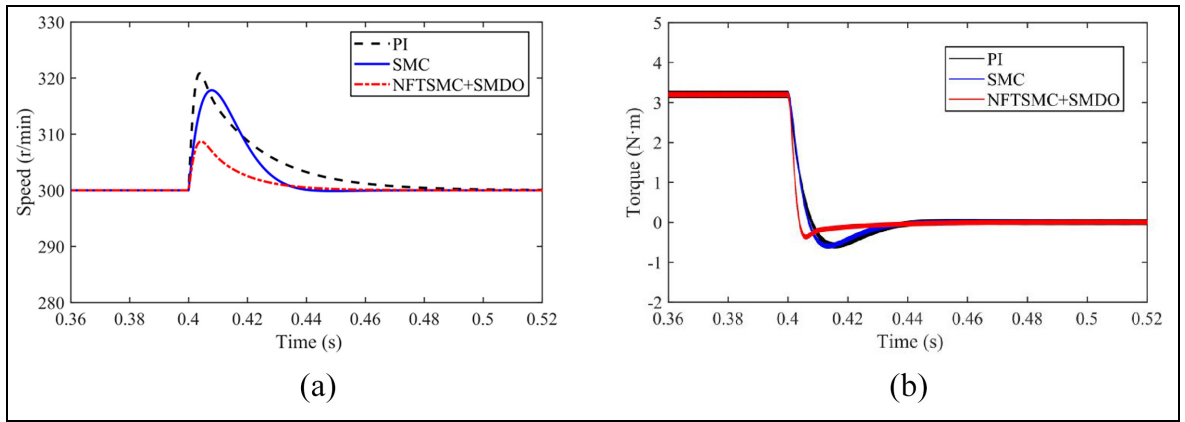


Figure 11. Simulation results of sudden load reduction at 300 r/min. (a) Speed. (b) Torque.

decreases at 300 r/min. Load torque decreases from 3.2 to 0 N·m at 0.4 s. Figure 11(a) shows that when the load is suddenly reduced, compared with PI and SMC, the method proposed in this article has the smallest speed increase of 9 r/min and the shortest recovery time to the given speed of 30 ms. From Figure 11(b), it can be seen that compared to PI and SMC, the method proposed in this article has a faster torque response and minimal torque fluctuation when the load suddenly decreases. Compared with Figure 9, the method proposed in this article has a smaller speed increase and shorter recovery time at 300 r/min. The simulation results show that the control strategy proposed in this article has a faster response and better immunity even at low speeds.

Therefore, from the above analysis, it can be concluded that when the load of the PMSM suddenly increases or decreases, the sliding mode disturbance observer can quickly and accurately estimate the load disturbance. The NFTSMC based on disturbance compensation adopted in this article has better anti-interference ability and faster response speed.

Experiments and analysis

In order to further verify the effectiveness of the control strategy, the load mutation experiment is carried out on the AGV with PMSM as the driving motor. The experimental platform of the AGV is shown in Figure 12. The AGV uses the STM32F28335 chip as the central controller, and the control strategy is downloaded to the central controller through Code Composer Studio 9.3 (CCS9.3) to achieve control of different algorithms. The driving mode of the AGV is the four-wheel independent drive, and the four driving motors adopt 750-W PMSM. The configuration and working characteristics of the four motors are the same. The encoder inside the PMSM can accurately obtain the motor's speed and position signals and provide real-time feedback to the upper computer. The upper computer is connected to the base station through Wi-Fi to achieve data collection and processing.

The AGV has an excellent experimental environment and no interference except for sudden load changes. When conducting load mutation experiments, the load of each mutation is at the center of gravity of

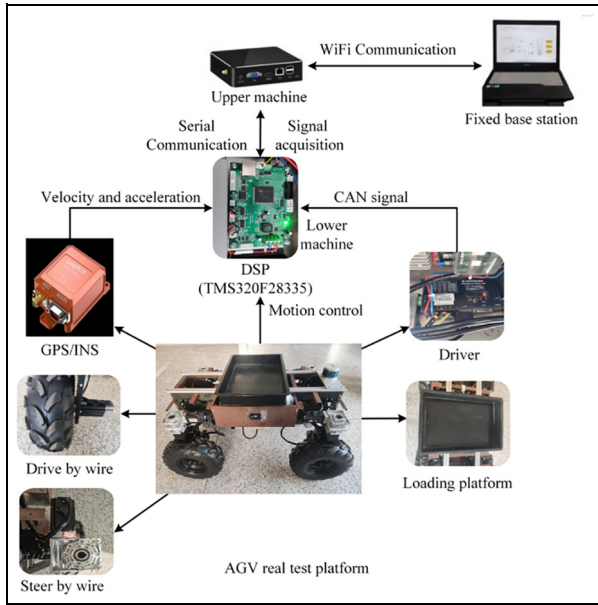


Figure 12. AGV experimental platform.

the AGV, ensuring that each motor is subjected to the same load. Because the AGV itself has mass, the load of the PMSM is not 0 when it is not loaded. The rated load of the AGV is 100 kg.

Figure 13 shows the speed response of the PMSM during start-up under three algorithmic controls, with a given speed of 800 r/min. As shown in the figure, Figure 13(a) shows PI control, with an overshoot of 27.5% at start-up and a recovery to a given speed of 69 ms.

Figure 13(b) is SMC control, with an overshoot of 26.8% at start-up and a recovery to a given speed of 43 ms. Figure 13(c) shows NFTSMC + SMDO control, the overshoot is 4.5% at start-up, and the recovery to the given speed is 19 ms. By comparison, the control strategy proposed in this article has the smallest overshoot and the fastest response speed during start-up.

In order to verify the anti-interference performance of different control strategies, load mutation experiments were conducted on AGV. The load of the AGV suddenly increases or decreases by 50 kg, and the given speed is 800 r/min. In order to further verify the anti-interference ability at low speeds, the control performance of different control methods for the PMSM at 300 r/min was compared. The load mutation experiment on AGV is shown in Figure 14. Figures 15–20 show the speed response and load disturbance estimation results of PMSM under different control strategies.

Figure 15 shows the speed response of PMSM under different control strategies when the load on the AGV suddenly increases by 50 kg. When the speed of the PMSM runs stably at 800 r/min, the load of the AGV suddenly increases from no-load to 50 kg. Figure 15(a)–(c) are the speed response results of PI, SMC, and NFTSMC + SMDO control, respectively. Figure 17 shows the load observation results of the SMDO. It can be seen from the figure that the SMDO can quickly and accurately observe the load disturbance. Figure 15 shows that the speed decrease during PI control is 34 r/min, and the speed recovery time is 0.51 s. During SMC control, the speed drops to 30 r/min, and the speed recovery time is 0.49 s. When NFTSMC + SMDO

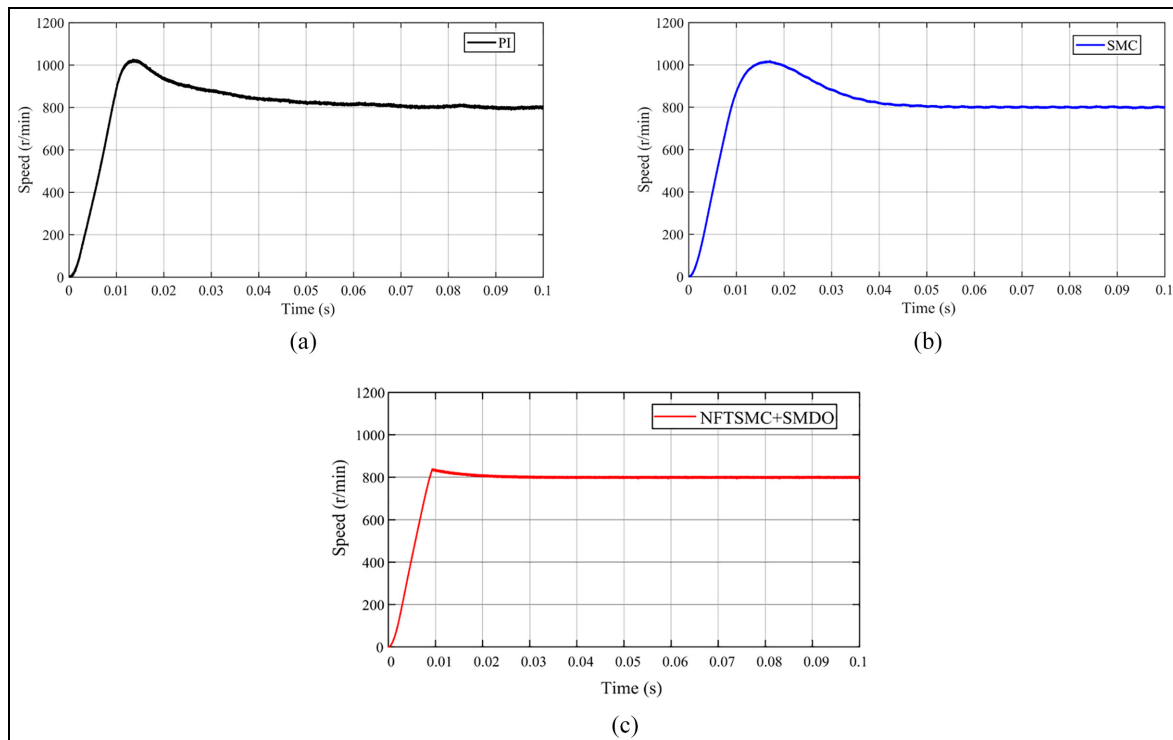


Figure 13. Speed response of PMSM under three algorithms. (a) PI. (b) SMC. (c) NFTSMC + SMDO.

control is used, the speed drops to 17 r/min, and the speed recovery time is 0.38 s. Compared with PI and SMC control, the speed drop of the control strategy proposed in this article is reduced by 2.12% and 1.63%, and the recovery time to the given speed is reduced by 25.4% and 22.4%, respectively. Therefore, the control strategy proposed in this article has a faster response speed and better anti-disturbance ability when the load suddenly increases.

Figure 16 shows the speed response of PMSM under a sudden increase of different control methods at 300 r/min. Figure 17 shows the load estimation results of the SMDO. From Figure 16(a), when the load suddenly increases, the speed under PI control decreases to 25 r/min, and the recovery time to the given speed is 0.49 s. From Figure 16(b), it can be seen that the speed decreased by 22 r/min during SMC control, and the time to recover to the given speed was 0.41 s. From Figure 16(c), it can be seen that the speed of NFTSMC + SMDO decreased by 10 r/min, and the time to recover to the given speed was 0.36 s. Compared with PI and SMC, the control method proposed in this article has a faster speed response and better anti-interference ability when the load suddenly increases. Table 2 compares the experimental results at 800 and 300 r/min. The data in the table shows that the control strategy proposed in this article also has good control performance at low speeds.

Figure 18 shows the speed response of PMSM under different control strategies when a load of the AGV



Figure 14. Load mutation experiment on AGV.

suddenly decreases by 50 kg. When the PMSM runs stably at 800 r/min, the load of the AGV suddenly decreases from 50 kg to 0. Figure 20 shows the load estimation results of the SMDO. It can be seen from the figure that the SMDO can quickly and accurately observe the load disturbance. Figure 18(a)–(c) shows the speed response results under PI, SMC, and NFTSMC + SMDO control, respectively. From Figure 18, it can be seen that the speed increase during PI control is 36 r/min, and the speed recovery time is 0.49 s. During SMC control, the speed increases to 32 r/min, and the speed recovery time is 0.48 s. During

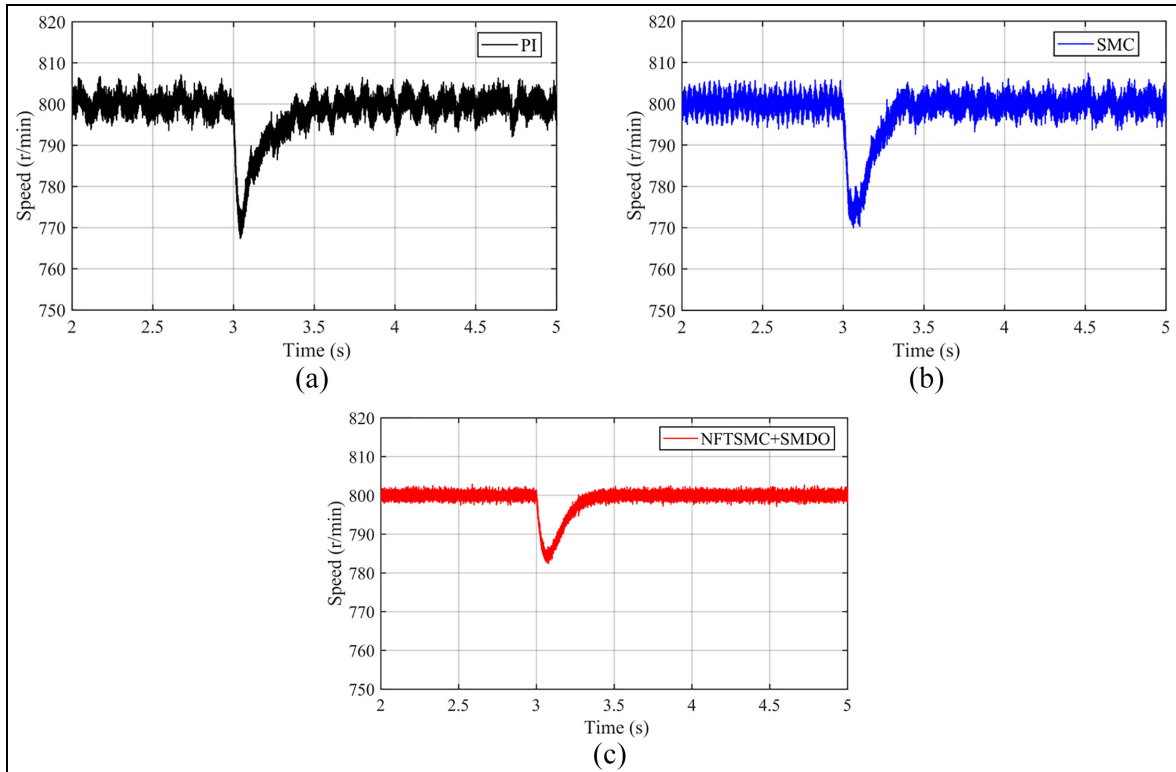


Figure 15. Speed response of PMSM under sudden load increase at 300 r/min. (a) PI. (b) SMC. (c) NFTSMC + SMDO.

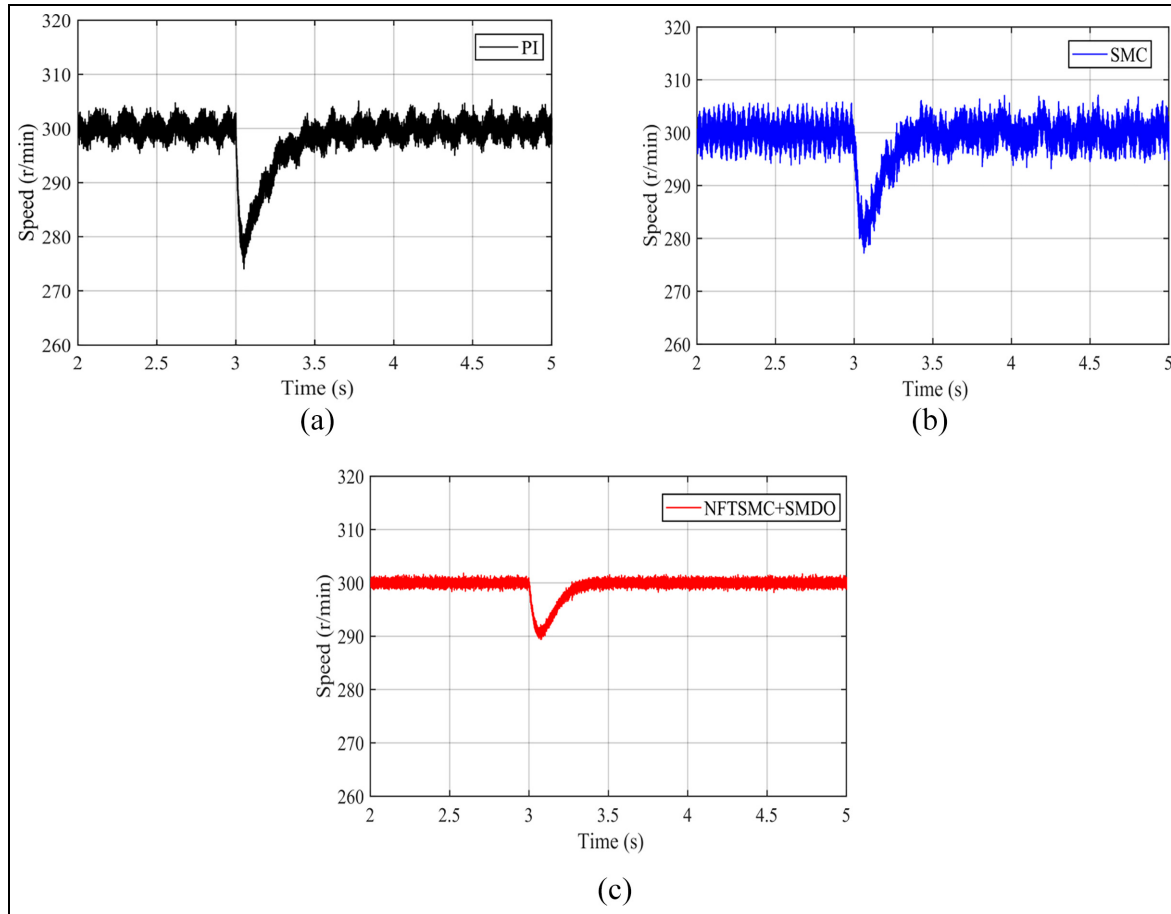


Figure 16. Speed response of PMSM under sudden load increase at 300 r/min. (a) PI. (b) SMC. (c) NFTSMC + SMDO.

NFTSMC + SMDO control, the speed increases to 18 r/min, and the speed recovery time is 0.44 s. Compared with PI and SMC control, the speed of the control strategy proposed in this article increases by 2.25% and 1.75%, and the time to return to the given speed decreases by 10.2% and 8.3%, respectively. Therefore, the control strategy proposed in this article has a faster response speed and better anti-disturbance ability when the load suddenly decreases.

Figure 19 shows the speed response of PMSM under sudden load reduction using different control methods at 300 r/min. Figure 20 shows the load estimation results of the SMDO. Figure 19(a) shows that when the load suddenly decreases, the speed under PI control increases to 25 r/min, and the time to recover to the given speed is 0.48 s. Figure 19(b) shows that the speed increases by 22 r/min during SMC control, and the time to recover to the given speed was 0.46 s. Figure 19(c) shows that under NFTSMC + SMDO control, the speed increases by 12 r/min, and the time to recover to the given speed is 0.41 s. Compared with PI and SMC, the control strategy proposed in this article has a faster speed response and better anti-interference ability when the load suddenly decreases. Table 3 compares the

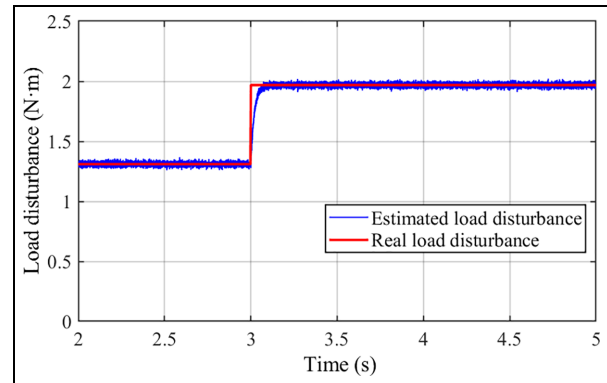


Figure 17. Disturbance estimation of SMDO in the case of sudden load increase.

experimental results of sudden load reduction at 800 and 300 r/min, indicating that the control strategy proposed in this article also has good control performance at low speeds.

From the simulation and experimental results, the control strategy proposed in this article improves the anti-interference ability and response speed of the PMSM speed control system.

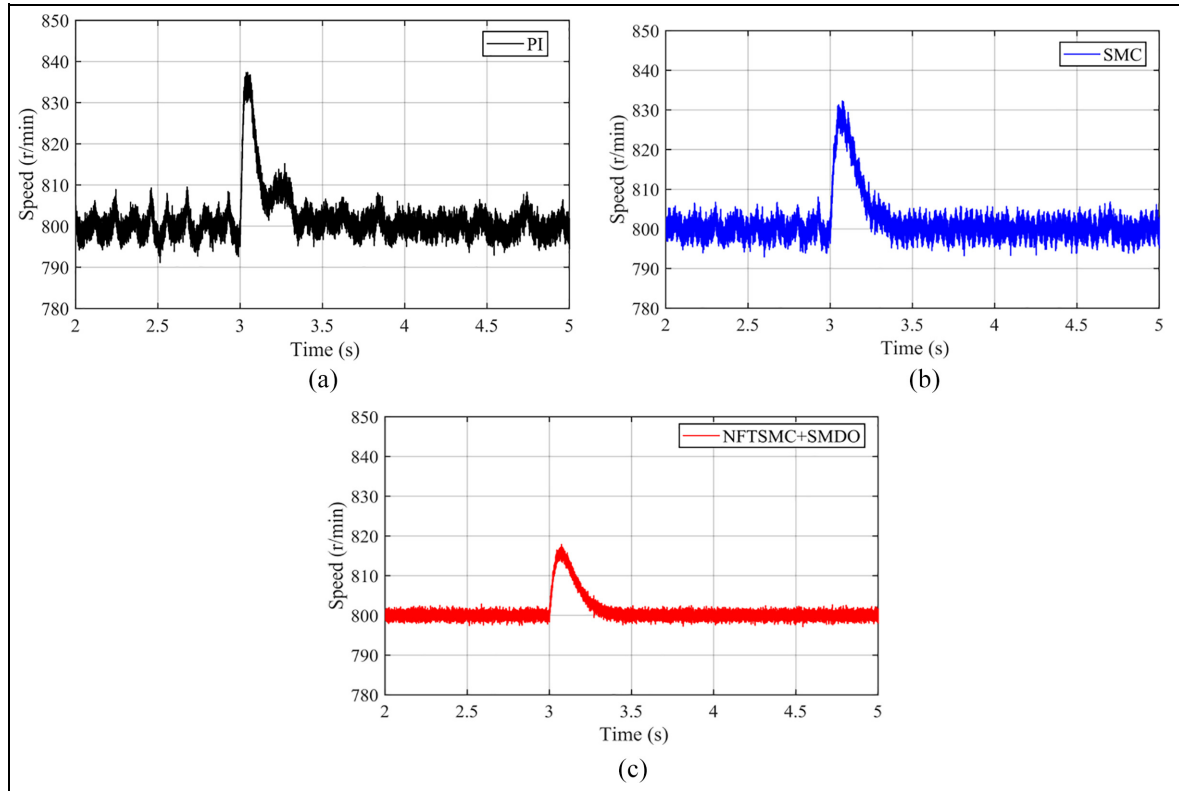


Figure 18. Speed response of PMSM under sudden load reduction at 800 r/min. (a) PI. (b) SMC. (c) NFTSMC + SMDO.

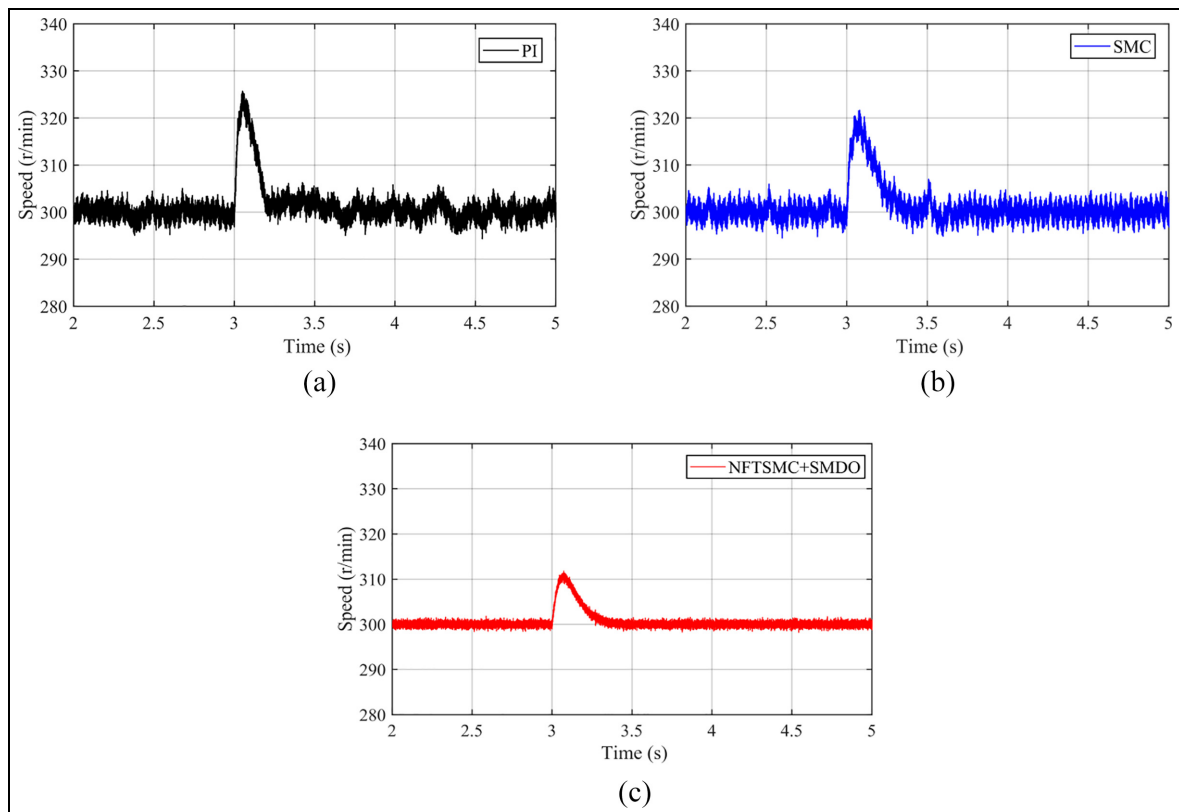


Figure 19. The speed response of PMSM under sudden load reduction at 300 r/min. (a) PI. (b) SMC. (c) NFTSMC + SMDO.

Table 2. Comparison of experimental results when the load increases.

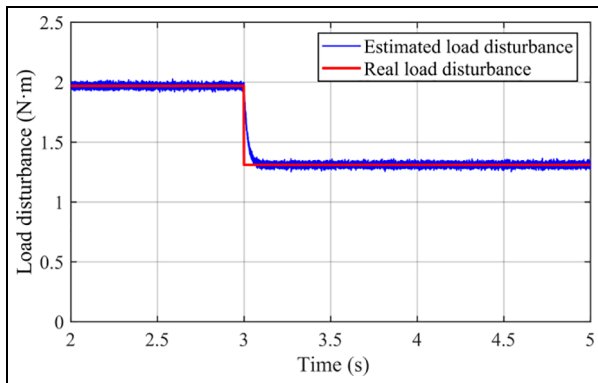
	Control scheme	Index	
		Speed decrease (r/min)	Recovery time(s)
300 r/min	PI	25	0.49
	SMC	22	0.41
	NFTSMC + SMDO	10	0.36
800 r/min	PI	34	0.51
	SMC	30	0.49
	NFTSMC + SMDO	17	0.38

PI: proportional-integral; SMC: sliding mode control; NFTSMC: non-singular fast terminal sliding mode control; SMDO: sliding mode disturbance observer.

Table 3. Comparison of experimental results when the load decreases.

	Control scheme	Index	
		Speed decrease (r/min)	Recovery time (s)
300 r/min	PI	26	0.48
	SMC	22	0.46
	NFTSMC + SMDO	12	0.41
800 r/min	PI	36	0.49
	SMC	32	0.48
	NFTSMC + SMDO	18	0.44

PI: proportional-integral; SMC: sliding mode control; NFTSMC: non-singular fast terminal sliding mode control; SMDO: sliding mode disturbance observer.

**Figure 20.** Disturbance estimation of SMDO in the case of sudden load decrease.

Conclusion

In this article, an NFTSMC based on perturbation compensation is proposed for the problem that PI control cannot effectively inhibit the sudden change in speed of PMSM when the load changes suddenly. First, an NFTSMC based on a new-reaching law is proposed. The new reaching law not only weakens the system chattering problem but also can adaptively adjust the approach speed according to the system's state. Compared with traditional sliding mode control, it can effectively suppress sliding mode chattering and improve response speed. Then, in order to improve the adverse effects of external load disturbances on the system, an SMDO is used to estimate the load disturbance, and the disturbance estimation value is

feed-forward compensated to the NFTSMC. Finally, a composite control method combining NFTSMC and SMDO is developed to improve the immunity performance of PMSM. The simulation and experimental results show that the control strategy proposed in this article can achieve satisfactory performance, with a fast transient response and good anti-interference ability. It is further demonstrated that the control strategy proposed in this article is able to improve the speed stability of AGV when sudden load changes. Then, further research work is still needed in the future because the effect of the variation of the internal parameters of the PMSM on the rotational speed was not considered. In order to solve this problem, we will investigate additional techniques to recognize parameter variations and make real-time corrections.

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Declaration of conflicting interests

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References

1. Liu W, Zhang QJ, Wan YD, et al. Yaw stability control of automated guided vehicle under the condition of centroid variation. *J Braz Soc Mech Sci Eng* 2021; 44(1): 18.
2. Liu W, Jing C, Wan P, et al. Combining extended Kalman filtering and rapidly-exploring random tree: an improved autonomous navigation strategy for four-wheel steering vehicle in narrow indoor environments. *Proc IMechE, Part I: J Syst Control Eng* 2022; 236(5): 883–896.
3. Wang XJ, Chen Y, Lu YL, et al. Dynamic surface method-based adaptive backstepping control for the permanent magnet synchronous motor on parameter identification. *Proc IMechE, Part I: J Syst Control Eng* 2019; 233(9): 1172–1181.
4. Jiang Q, Liu JP, Yu JP, et al. Full state constraints and command filtering-based adaptive fuzzy control for permanent magnet synchronous motor stochastic systems. *Inf Sci* 2021; 567: 298–311.
5. Jon R, Wang ZS, Luo CM, et al. Adaptive robust speed control based on recurrent Elman neural network for sensorless PMSM servo drives. *Neurocomputing* 2016; 227: 131–141.
6. El-Sousy F. Adaptive hybrid control system using a recurrent RBFN-based self-evolving fuzzy-neural-network for PMSM servo drives. *Appl Soft Comput J* 2014; 21: 509–532.
7. Feng XP, Xie SR, Zhang ZY, et al. Research on speed loop control of IPMSM based on Fuzzy linear active disturbance rejection control. *Energy Rep* 2022; 8: 804–812.
8. Zhang XG and Wang ZW. Simple robust model predictive current control for PMSM drives without flux-linkage parameter. *IEEE Trans Ind Electron* 2023; 70(4): 3515–3524.
9. Zhu PK, Chen Y and Li M. Terminal sliding mode control of permanent magnet synchronous motor based on the reaching law. *Proc IMechE, Part I: J Syst Control Eng* 2020; 234(7): 849–859.
10. Xu W, Junejo AK, Liu Y, et al. Improved continuous fast terminal sliding mode control with extended state observer for speed regulation of PMSM drive system. *IEEE Trans Veh Technol* 2019; 68(11): 10465–10476.
11. Wang Y, Yu HT, Che ZY, et al. The direct speed control of PMSM based on terminal sliding mode and finite time observer. *Processes* 2019; 7(9): 624.
12. Liu W, Chen SY and Huang HX. Adaptive nonsingular fast terminal sliding mode control for permanent magnet synchronous motor based on disturbance observer. *IEEE Access* 2019; 7: 153791–153798.
13. Wang AM and Wei SJ. Sliding mode control for permanent magnet synchronous motor drive based on an improved exponential reaching law. *IEEE Access* 2019; 7: 146866–146875.
14. Xu B, Zhang L and Ji W. Improved non-singular fast terminal sliding mode control with disturbance observer for PMSM drives. *IEEE Trans Transp Electrification* 2021; 7(4): 2753–2762.
15. Qu SC, Xu WJ, Zhao JH, et al. Design and implementation of a fast sliding-mode speed controller with disturbance compensation for SPMSM system. *IEEE Trans Transp Electrification* 2021; 7(4): 2611–2622.
16. Sun XD, Cao JH, Lei G, et al. A composite sliding mode control for SPMSM drives based on a new hybrid reaching law with disturbance compensation. *IEEE Trans Transp Electrification* 2021; 7(3): 1427–1436.
17. Lu E, Li W, Yang XF, et al. Anti-disturbance speed control of low-speed high-torque PMSM based on second-order non-singular terminal sliding mode load observer. *ISA Trans* 2019; 88: 142–152.
18. Zhang XG, Sun LZ, Zhao K, et al. Nonlinear speed control for PMSM system using sliding-mode control and disturbance compensation techniques. *IEEE Trans Power Electron* 2013; 28(3): 1358–1365.
19. Nguyen AT, Basit BA, Choi HH, et al. Disturbance attenuation for surface-mounted PMSM drives using nonlinear disturbance observer-based sliding mode control. *IEEE Access* 2020; 8: 86345–86356.
20. Wu YJ and Li GF. Adaptive disturbance compensation finite control set optimal control for PMSM systems based on sliding mode extended state observer. *Mech Syst Signal Process* 2018; 98: 402–414.
21. Dai YC, Ni SF, Xu DZ, et al. Disturbance-observer based prescribed-performance fuzzy sliding mode control for PMSM in electric vehicles. *Eng Appl Artif Intell* 2021; 104: 104361.
22. Yang T, Deng YT, Li HW, et al. Fast integral terminal sliding mode control with a novel disturbance observer based on iterative learning for speed control of PMSM. *ISA Trans* 2023; 134: 460–471.

Appendix I

Notation

B	viscous friction coefficient
i_d, i_q	d-axis and q-axis reference current
J	moment of inertia
L_d, L_q	d-axis and q-axis stator inductance
p_n	number of the pole pairs
R	stator resistance
T_e	electromagnetic torque
T_L	load torque
T_L^*	observed load disturbance
s	speed controller sliding surface

s_1	sliding mode observer sliding mode surface	ω_e	electrical angular velocity
u_d, u_q	d -axis and q -axis voltage	ω	reference speed
		ω_m	mechanical angular velocity
ψ_f	flux linkage of the rotor permanent magnet	ω_m^*	estimated angular velocity